EXPERIMENT 1: FIND-S ALGORITHM

EXPERIMENT QUESTION: IMPLEMENT AND DEMONSTRATE THE FIND-S ALGORITHM FOR FINDING THE MOST SPECIFIC HYPOTHESIS BASED ON A GIVEN SET OF TRAINING DATA SAMPLES.

Subproblem 1.1: Enjoy Sport Prediction

1. Aim

To implement FIND-S and learn the most specific hypothesis for predicting whether a person enjoys a sport from weather conditions.

2. Dataset Description

The training examples are shown below. The last column is the target class.

Sky	Temperature	Humidity	Wind	Enjoy Sport
Sunny	Warm	Normal	Strong	Yes
Sunny	Warm	High	Strong	Yes
Rainy	Cold	High	Strong	No
Sunny	Warm	High	Weak	Yes

3. Algorithm

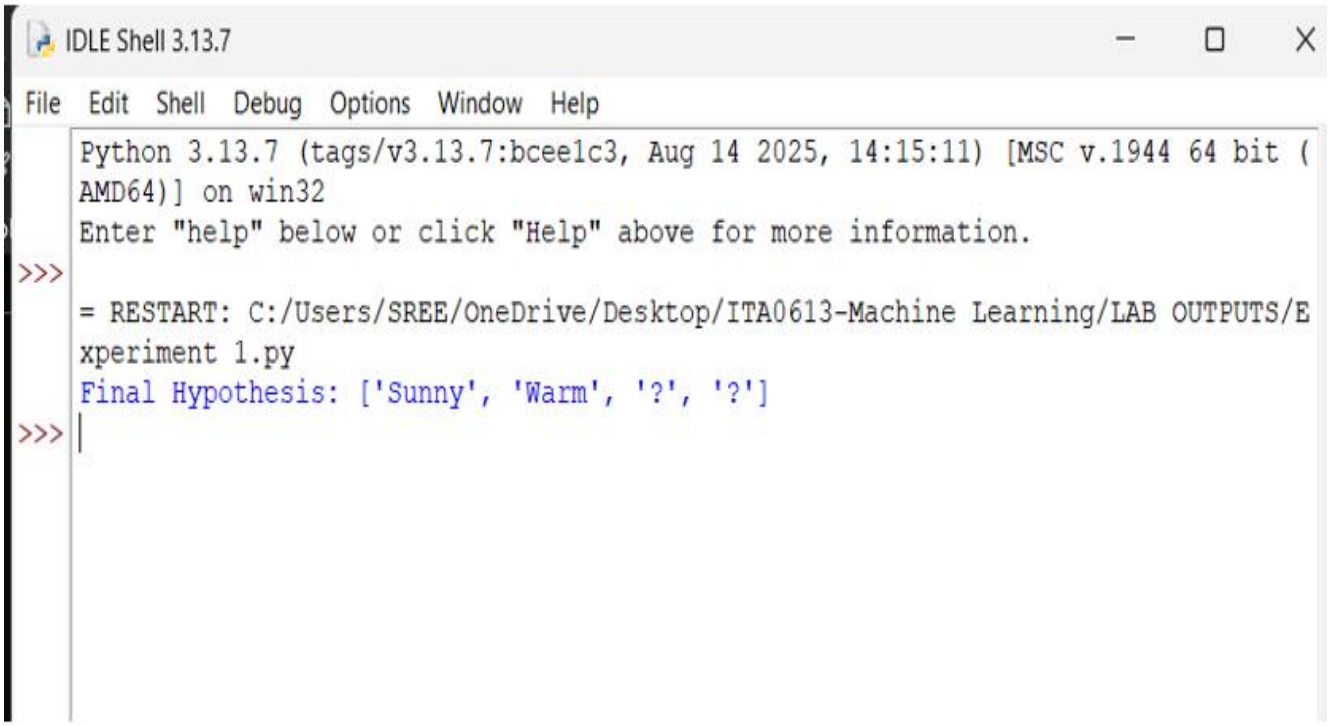
1. Initialize H to the most specific hypothesis (Ø in every attribute).
2. Read each training example from left to right.
3. If the target is No, ignore the example.
4. If the target is Yes, compare each attribute with H.
5. Replace Ø with the example value; if a value conflicts with the current specific value, replace it with ?; otherwise keep it unchanged.
6. Display the final hypothesis after all examples have been processed.

PROGRAM

```
data = [
    ['Sunny', 'Warm', 'Normal', 'Strong', 'Yes'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Yes'],
```

```
    ['Rainy', 'Cold', 'High', 'Strong', 'No'],
    ['Sunny', 'Warm', 'High', 'Weak', 'Yes']
]
h = ['Ø'] * 4
for row in data:
    if row[-1] == 'Yes':
        for i in range(4):
            if h[i] == 'Ø':
                h[i] = row[i]
            elif h[i] != row[i]:
                h[i] = '?'
print("Final Hypothesis:", h)
```

OUTPUT



Subproblem 1.2: Weather Suitability Prediction

1. Aim

To apply FIND-S to learn the common conditions in examples labelled Suitable Weather = Yes.

2. Dataset Description

The training examples are shown below. The last column is the target class.

Sky	Temperature	Humidity	Wind	Suitable Weather
Sunny	Hot	High	Weak	Yes
Sunny	Hot	High	Strong	Yes
Rainy	Cold	High	Weak	No
Sunny	Mild	Normal	Weak	Yes

3. Algorithm

1)Initialize H to the most specific hypothesis (Ø in every attribute).

2)Read each training example from left to rightIf the target is No, ignore the example.

3)If the target is Yes, compare each attribute with H.

4)Replace Ø with the example value; if a value conflicts with the current specific value, replace it with?; otherwise keep it unchanged.

5)Display the final hypothesis after all examples have been processed

PROGRAM

```
data = [
    ['Sunny', 'Hot', 'High', 'Weak', 'Yes'],
    ['Sunny', 'Hot', 'High', 'Strong', 'Yes'],
    ['Rainy', 'Cold', 'High', 'Weak', 'No'],
    ['Sunny', 'Mild', 'Normal', 'Weak', 'Yes']
]
h = ['Ø'] * 4
```

```
for row in data:
    if row[-1] == 'Yes':
        for i in range(4):
            if h[i] == 'Ø':
                h[i] = row[i]
            elif h[i] != row[i]:
                h[i] = '?'
print("Final Hypothesis:", h)
```

OUTPUT

```
>>>
= RESTART: C:/Users/SREE/OneDrive/Desktop/ITA0613-Machine Learning/LAB OUTPUTS/E
xperiment 1.py
Final Hypothesis: ['Sunny', '?', '?', '?']
>>>
```

Subproblem 1.3: Student Performance Prediction

1. Aim

To use FIND-S to identify the attribute pattern shared by students whose result is Pass.

2. Dataset Description

The training examples are shown below. The last column is the target class.

Attendance	Internal Marks	Assignment	Result
High	Good	Regular	Yes
High	Good	Irregular	Yes
Low	Poor	Regular	No
Medium	Good	Regular	Yes

3. Algorithm

1)Initialize H to the most specific hypothesis (Ø in every attribute).

2)Read each training example from left to right.

3)If the target is No, ignore the example.

4)If the target is Yes, compare each attribute with H.

5)Replace Ø with the example value; if a value conflicts with the current specific value, replace it with ?; otherwise keep it unchanged.

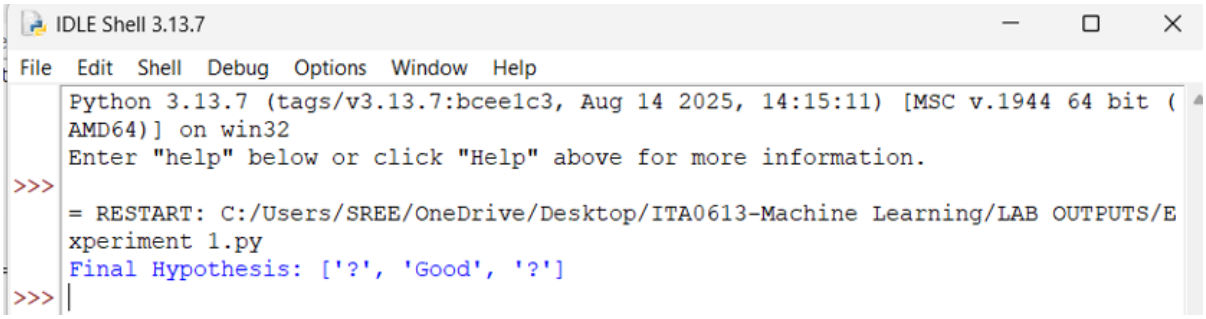
6)Display the final hypothesis after all examples have been processed.

PROGRAM

```
data = [
    ['High', 'Good', 'Regular', 'Yes'],
    ['High', 'Good', 'Irregular', 'Yes'],
    ['Low', 'Poor', 'Regular', 'No'],
    ['Medium', 'Good', 'Regular', 'Yes']
]

h = ['Ø'] * 3
for row in data:
    if row[-1] == 'Yes':
        for i in range(3):
            if h[i] == 'Ø':
                h[i] = row[i]
            elif h[i] != row[i]:
                h[i] = '?'
print("Final Hypothesis:", h)
```

OUTPUT



Subproblem 1.4: Animal Classification

1. Aim

To implement FIND-S to learn the common attribute pattern of animals belonging to a target category.

2. Dataset Description

The training examples are shown below. The last column is the target class.

Body Temperature	Legs	Habitat	Animal Category
Warm	Four	Land	Yes
Warm	Four	Land	Yes
Cold	Two	Water	No
Warm	Four	Water	Yes

3. Algorithm

1)Initialize H to the most specific hypothesis (Ø in every attribute).

2)Read each training example from left to right.

3)If the target is No, ignore the example.

4)If the target is Yes, compare each attribute with H.

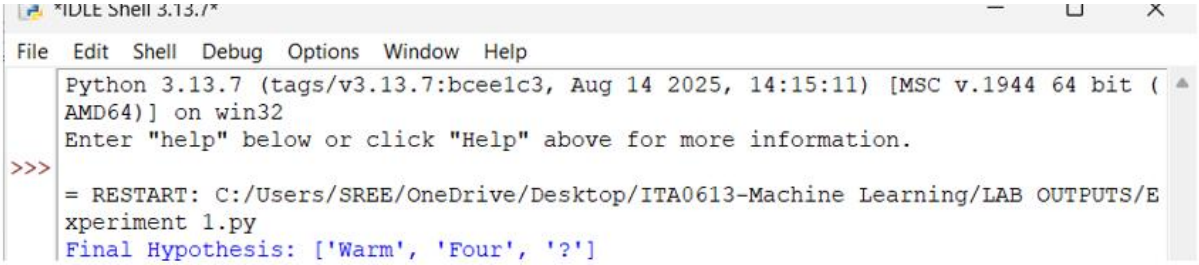
5)Replace Ø with the example value; if a value conflicts with the current specific value, replace it with ?; otherwise keep it unchanged.

6)Display the final hypothesis after all examples have been processed.

PROGRAM

```
data = [
    ['Warm', 'Four', 'Land', 'Yes'],
    ['Warm', 'Four', 'Land', 'Yes'],
    ['Cold', 'Two', 'Water', 'No'],
    ['Warm', 'Four', 'Water', 'Yes']
]
h = ['Ø'] * 3
for row in data:
    if row[-1] == 'Yes':
        for i in range(3):
            if h[i] == 'Ø':
                h[i] = row[i]
            elif h[i] != row[i]:
                h[i] = '?'
print("Final Hypothesis:", h)
```

OUTPUT



Subproblem 1.5: Computer Purchase Prediction

1. Aim

To apply FIND-S to learn the common attributes of computers labelled as purchased.

2. Dataset Description

The training examples are shown below. The last column is the target class.

Price	Speed	Memory	Purchase
High	Fast	Large	Yes
High	Fast	Small	Yes
Low	Slow	Small	No
Medium	Fast	Large	Yes

3. Algorithm

1)Initialize H to the most specific hypothesis (Ø in every attribute).

2)Read each training example from left to right.

3)If the target is No, ignore the example.

4)If the target is Yes, compare each attribute with H.

5)Replace Ø with the example value; if a value conflicts with the current specific value, replace it with ?; otherwise keep it unchanged.

6)Display the final hypothesis after all examples have been processed.

PROGRAM

```
data = [
    ['High', 'Fast', 'Large', 'Yes'],
```

```
    ['High', 'Fast', 'Small', 'Yes'],
    ['Low', 'Slow', 'Small', 'No'],
    ['Medium', 'Fast', 'Large', 'Yes']
]
h = ['Ø'] * 3
for row in data:
    if row[-1] == 'Yes':
        for i in range(3):
            if h[i] == 'Ø':
                h[i] = row[i]
            elif h[i] != row[i]:
                h[i] = '?'
print("Final Hypothesis:", h)
```

OUTPUT

